DATA PIPELINE PROJECT · SPICED ACADEMY MLE

Did congestion pricing change NYC's Green Taxis?

An ELT pipeline with Postgres, dbt and Prefect

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The brief, and the question I added

COURSE BRIEF

Revenue per day for Green Taxis, January to March 2025

Download the data, build an ETL or ELT pipeline, add it up per day. Bonus: run it with Prefect.

MY QUESTION

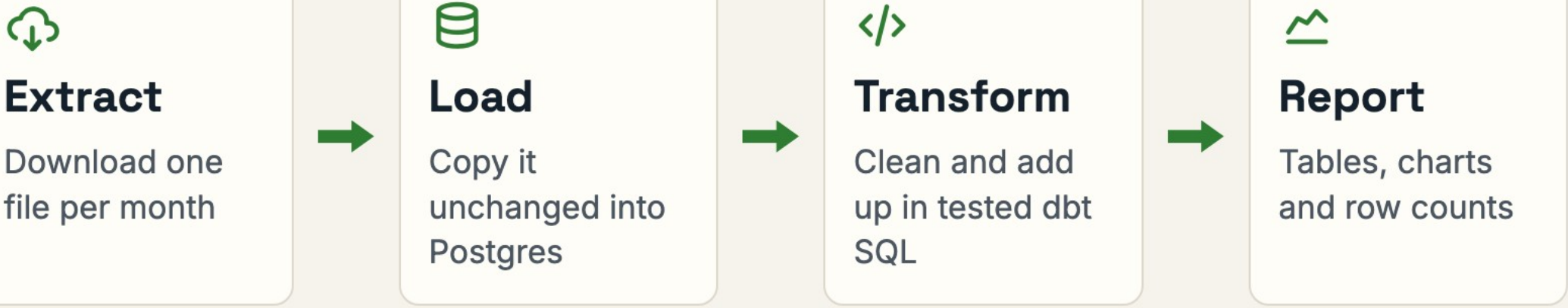
Did congestion pricing change trips and revenue?

Since 5 January 2025, taxi trips into lower Manhattan pay $0.75. I extended the pipeline to 18 months to compare two winters.



How it works: four steps, run by Prefect

Prefect · retries · logs · results in a web UI · a monthly schedule



Why ELT, not ETL? The raw data is stored first and cleaned in SQL afterwards. Change a rule and re-run: no new download, and every row traces back to its file.

What the data needed before adding it up

146,486
rows downloaded

→ **146,475**
rows kept

→ **90**
days, one row each

11 trips dated outside the quarter

December 2024 and 1 April. Left in, the result would show 94 days instead of 90.

478 refunds: −$6,462.61

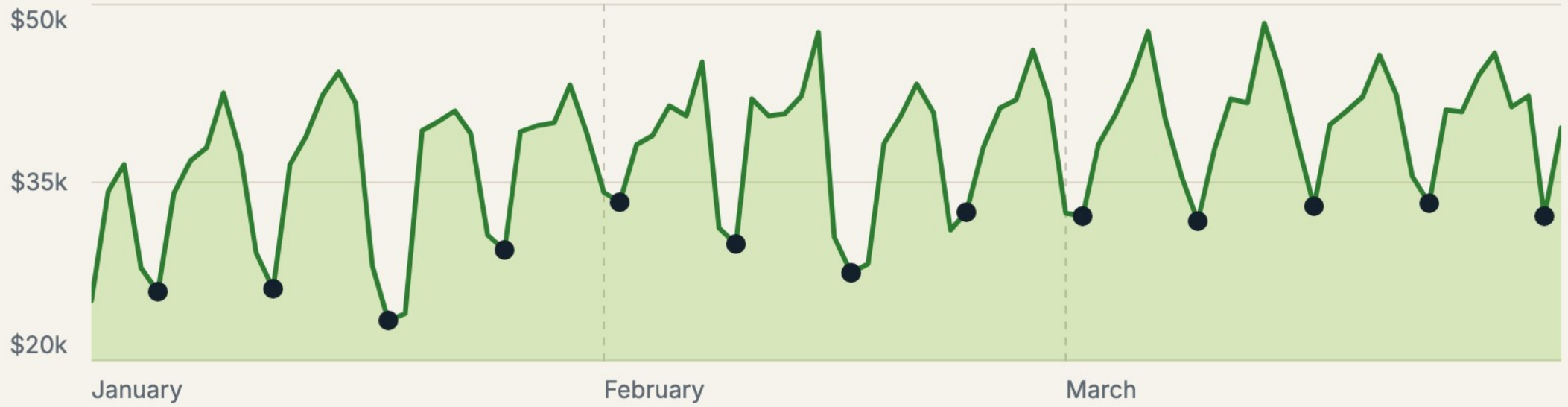
Negative amounts. Counted apart from trips, as gross revenue, refunds and net revenue.

Course result: $3.4 million in 90 days

$3,395,290.56
net revenue, January to March 2025

$37,725
on an average day

+45%
Thursdays over Sundays

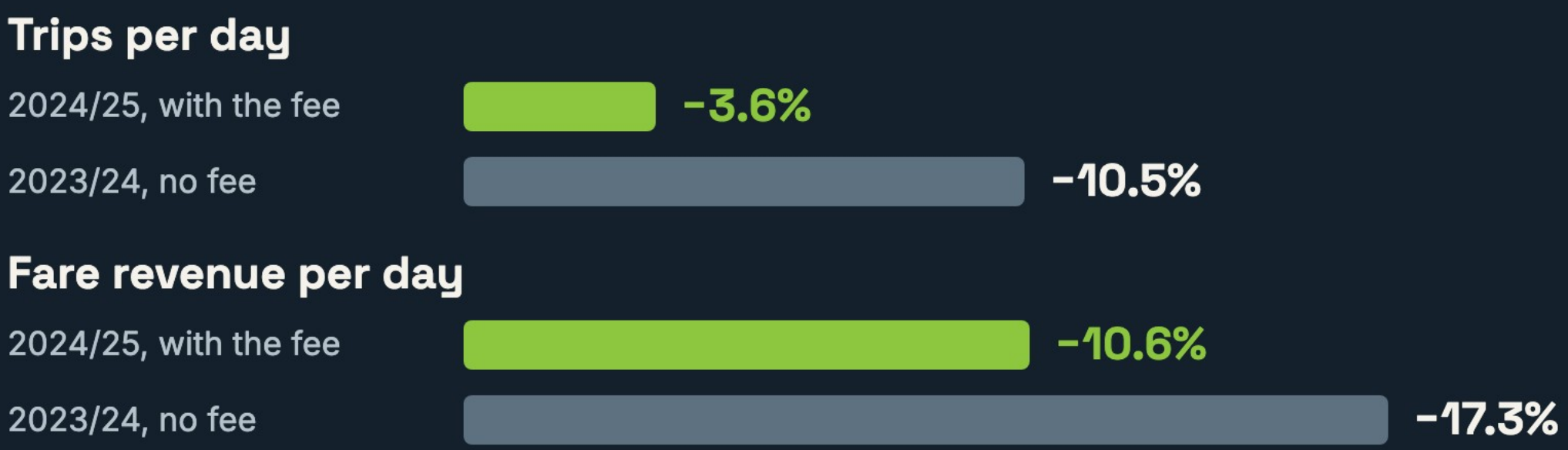


Green line: net revenue per day · Dark dots: Sundays, the quietest day of every week

Same result as a classmate's independent pipeline, to the cent

Congestion pricing: no visible effect

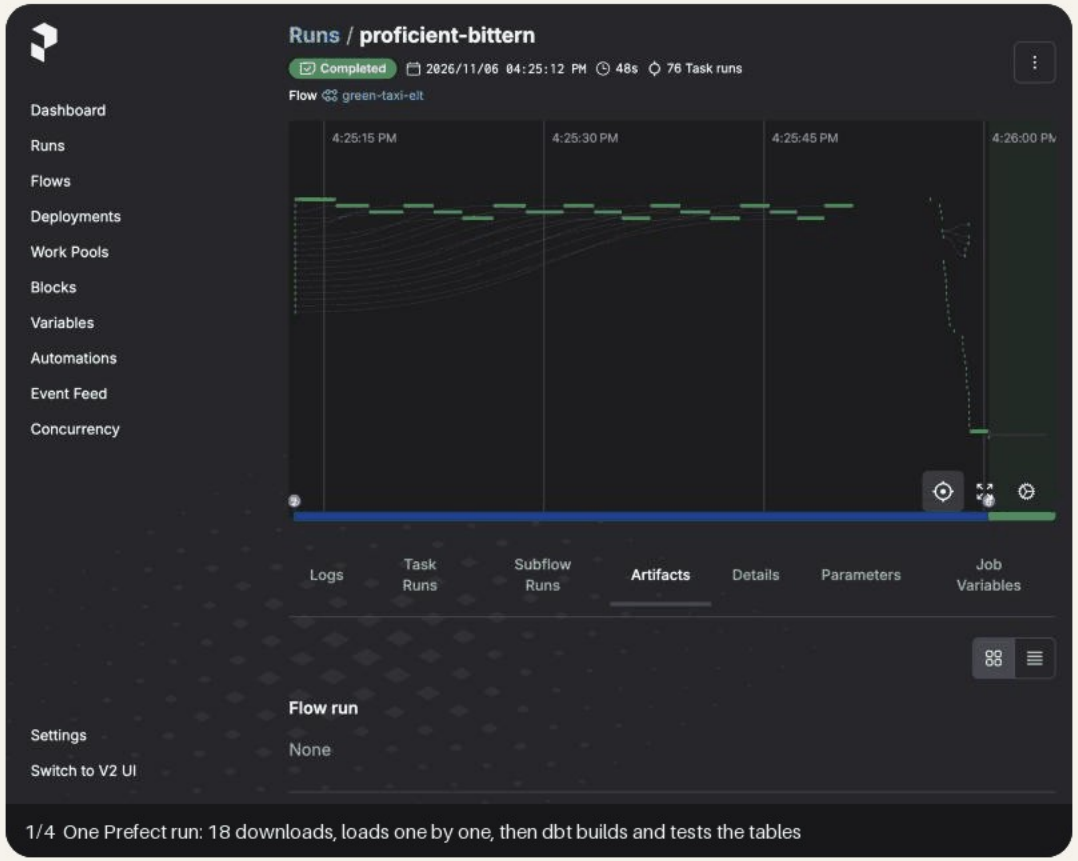
Weekdays, change after 5 January against the same dates one year earlier



The drop after 5 January is the usual winter dip, and it was smaller with the fee than without.

About 1 in 12 weekday trips paid the $0.75 fee

Built so the numbers can be trusted



- **34 data checks on every run.** A wrong number stops the run before it reaches the output.
- **34 code tests,** run by GitHub on every pull request.
- **Safe to re-run.** Same input, same result, never duplicates.
- **Verified from a fresh clone,** the way a reviewer would run it.

Left: an 18-month run in the Prefect web UI

What I'd do next

Production-ready Prefect

Server and worker in Docker, so runs don't depend on my laptop

New months, automatically

Fetch each month as soon as the city publishes it

Revenue by borough

Break it down with the taxi-zone lookup

Thank you. Questions?

github.com/sean-mas/nyc-taxi-congestion-pricing-pipeline